

Labor Income Inequality and Stabilization Policies

JOB MARKET PAPER

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Abstract

We study how rising permanent labor-income inequality shapes monetary and fiscal transmission in a Heterogeneous-Agent New-Keynesian model with a non-homothetic taste for wealth. Higher inequality increases the share of hand-to-mouth households, raising aggregate MPCs and lowering the effective intertemporal elasticity of substitution. As a result, the direct effect of monetary policy weakens, while the indirect effect strengthens, leading to a larger output response. A higher share of capital gains generated by monetary policy are paid to wealthy households, making monetary policy more regressive. For fiscal policy, higher MPCs raise impact and cumulative multipliers and make deficit-financed expansions more likely to be self-financing. Finally, the fiscal response to a monetary policy shock becomes more important to determine the output response to a change in interest rate.

Keywords: Inequality, permanent income, monetary policy, fiscal policy, interest rates, heterogeneous agent model.

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Contents

1	A HANK Model with Permanent Labor Income Inequality	5
2	The Intertemporal Keynesian Cross and Permanent Labor Income Inequality	10
3	Empirical Fit of the Model	15
3.1	Calibration	15
3.2	The Taste for Wealth and the Wealth Distribution	17
4	Impact of Rising Permanent Labor Income Inequality on Stabilization Policies	19
4.1	Steady-State Changes	19
4.2	Monetary Policy	22
4.3	Fiscal Policy	26
4.4	Interaction between fiscal and monetary policy	28
A	Deriving the Intertemporal Keynesian Cross	34
A.1	Intertemporal Keynesian Cross following a monetary shock	34
A.2	Decomposition of the effect of a change in the permanent labor income distribution	35
B	Analytical Approximation of Pareto Tails in the Model	36

Introduction

Labor-income inequality in the United States has risen sharply over the past four decades, with the top one percent doubling its share of pre-tax labor income since 1980 (Piketty et al. 2018a, Saez & Zucman 2020). A growing empirical literature attributes this rise primarily to an increase in the permanent component of labor-income heterogeneity—that is, to widening gaps in lifetime earnings profiles across cohorts (DeBacker et al. 2011, Bloom et al. 2017, Braxton et al. 2021, Guvenen et al. 2022). This structural change raises an important question for stabilization policy: how does the distribution of permanent labor income shape the transmission of monetary and fiscal policy?

We address this question using a Heterogeneous-Agent New-Keynesian (HANK) model with rich household heterogeneity and an intertemporal Keynesian cross (Auclert et al. 2024). The starting point is a neutrality benchmark. Under standard homothetic preferences, consumption is linear in permanent income (Straub 2019), and the distribution of permanent income is irrelevant for aggregate responses to interest-rate or fiscal shocks. Permanent-income heterogeneity thus appears immaterial for stabilization policy, echoing earlier neutrality results.

This benchmark, however, conflicts with an observed empirical fact: households with higher permanent income save disproportionately more out of permanent income. Standard preferences cannot generate this pattern. We therefore depart from homotheticity by introducing a non-homothetic taste for wealth, following evidence in Kumhof et al. (2015), Michau (2018), Straub (2019), Mian et al. (2021), and others. Wealth acts as a luxury good, inducing concavity in consumption with respect to permanent income and breaking the neutrality result. Our model can also match the observed wealth distribution in the U.S., a realistic level of aggregate wealth, and an aggregate MPC of 0.5 – patterns that have been difficult to match for one-asset HANK models (Kaplan et al. 2018, Auclert et al.

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2024).

Once wealth enters the utility function in this way, permanent labor-income inequality becomes a key determinant of aggregate dynamics. A rise in permanent inequality increases the mass of hand-to-mouth households, raises the aggregate marginal propensity to consume, and lowers the aggregate elasticity of intertemporal substitution. Intuitively, greater dispersion in permanent income increases the equilibrium weight placed on the saving motive of wealth-rich households, depresses the real interest rate, and shortens households' effective planning horizons. Consumption therefore becomes more front-loaded, more sensitive to contemporaneous income, and less responsive to intertemporal price movements.

These shifts in consumer behavior substantially alter the transmission of stabilization policy. A higher level of permanent-income inequality weakens the direct intertemporal-substitution channel of monetary policy but strengthens indirect income-driven channels. Quantitatively, the latter dominate: monetary policy becomes more powerful, even though the standard Euler-equation channel attenuates. For fiscal policy, higher permanent-income inequality amplifies both impact and cumulative multipliers and increases the likelihood that deficit-financed transfers are self-financing in the sense of Angeletos et al. (2024).

Our model also implies that the redistributive effect of monetary policy becomes stronger when inequality is already high. A monetary policy shock generates capital gains that are paid mostly to wealthy households, significantly and persistently increasing the wealth shares of the top 1% and 0.1%. An last important implication is that fiscal–monetary interactions become more consequential when permanent-income inequality is high. Because a larger share of monetary transmission operates through household disposable income, the output response to a monetary shock depends more heavily on how the government adjusts its fiscal stance. When monetary easing expands fiscal space, consolidation dampens the output response while temporary tax cuts amplify it.

The remainder of the paper develops these results formally, quantifies them in a calibrated model, and discusses their implications for stabilization policy in unequal economies.

Literature. Our paper belongs to the old and large literature that investigates the transmission channels of monetary policy on aggregate variables. Over the last two decades, this literature has gradually moved away from a representative-agent framework to explore the interactions between distributions of income and wealth, and monetary policy. The wealth distribution matters for monetary policy as it determines the share of direct and indirect effects in the transmission of shocks. Micro survey data on household portfolio reveals that a large portion of households do not hold liquid wealth (Kaplan et al. 2014). Those households do not react to interest rate changes but are very sensitive to changes in their disposable income. When including that heterogeneity, Kaplan et al. (2018) shows that most of the transmission of monetary policy goes through indirect general equilibrium effects (mainly through the change in labor demand and in taxes). The composition of a monetary shock matters for policymakers for at least two reasons. First, if most of the effect of monetary policy goes through indirect effects, the fiscal response becomes key as the Ricardian equivalence fails to hold (Alves et al. 2020). Second, Holm et al. (2021) shows that indirect effects take time to materialize. As a result, if most of the effect of monetary policy goes through indirect effects, monetary policy is not capable of raising aggregate output in a short time frame.

Monetary policy has also a feedback effect on the wealth and the income distributions (Coibion et al. 2017, Mumtaz & Theophilopoulou 2017). Those papers show that contractionary monetary policy is usually procyclical while expansionary monetary policy tends to dampen income inequality. This feedback effect from the monetary policy on the income distribution not only matters from an equity standpoint but also since it is one of the transmission channels of monetary policy on aggregate variables. Indeed, by redistributing income from low-MPC households to high-MPC households, expansionary monetary policy is amplified compared to the representative-agent benchmark (Auclert 2019).

While the interactions between wealth distribution and monetary policy have been extensively studied, the way the distribution of labor income shapes monetary policy has remained partially under the radar. The main reason is that the drivers behind the in-

crease in labor income inequality were not well understood until recently. The macro implication of a rise in the variance of shocks over the lifetime is vastly different from the rise in the variance of initial outcomes. Both Bloom et al. (2017) and Braxton et al. (2021) show that temporary earning risks have declined. DeBacker et al. (2011) and Guvenen et al. (2022) show that most of the increase in labor income inequality has come from an increase in the variance of initial outcomes. Our contribution is to study the implications of that increase on the way monetary policy shocks are transmitted.

Our paper uses recent theoretical and methodological advances made by Auclert et al. (2024). They show that modern micro-founded heterogeneous-agent New-Keynesian models feature an "intertemporal Keynesian cross", characterized by the sequence-space Jacobian of the consumption function. Our paper extends this methodology to monetary policy, and we recover a similar intertemporal Keynesian cross for monetary policy shocks.

Our paper also belongs to the growing literature that uses a non-homothetic taste for wealth to solve various puzzles. Carroll (1998) proposes wealth as a source of social status to explain why rich households have such high saving rates compared to the rest of the population. Straub (2019) shows that non-homothetic preferences capture the saving behavior along the distribution of permanent income and explain how variations in permanent labor income inequality increase wealth inequality and decrease the equilibrium interest rate. Kumhof et al. (2015) also uses a non-homothetic taste for wealth to match the saving behavior of richer households, in a model that captures the accumulation of debt in the U.S. in the years preceding the Great Recession. In their literature review, De Nardi et al. (2016) also mention non-homothetic bequest motive to explain why a large number of households die with significant amounts of wealth. Within the New-Keynesian literature, Michau (2018) and Mian et al. (2021) also include bonds in the utility to study the over-accumulation of savings in modern economies. Lastly, in Gaillard & Wangner (2021), the non-homothecity in the taste for wealth allows to match the thicker tail of the income distribution compared to the consumption distribution. This finding is consistent with the concavity in the consumption function with respect to permanent income generated by this non-homothecity.

Our emphasis on the interactions between non-homothetic preference in the saving behavior and monetary policy is shared by Michaillat & Saez (2021) and Melcangi & Sterk (2020). Consistent with Michaillat & Saez (2021), the presence of bonds-in-the-utility solves the forward guidance puzzle, even in the absence of idiosyncratic shocks. Melcangi & Sterk (2020) is also concerned by the change in the policy channels and in the aggregate effect of monetary policy. The paper also finds out that, in the US since 1980, the power of monetary policy has strengthened following the rise in the stock market participation channel.

Section 1 presents our HANK model with permanent labor income inequality and a non-homothetic taste for wealth. Section 2 derives the Intertemporal Keynesian Cross (IKC) in our model and proposes a decomposition of the impact of rising permanent labor income inequality on this IKC. Section 3 describes the calibration and studies the empirical fit of the model. Section 4 presents our main quantitative results on the impact of rising permanent labor income inequality on monetary and fiscal policy.

1 A HANK Model with Permanent Labor Income Inequality

This section presents a canonical Heterogeneous-Agent New Keynesian (HANK) model with rigid wages, non-homothetic preferences for wealth, and permanent labor income inequality.

Households. The economy is composed of a continuum of households, who face idiosyncratic productivity shocks e_t , a constant death probability ζ , and are *ex ante* heterogeneous in their permanent productivity level z_i . Each permanent type i has a total mass of ω_i , with $\sum_i \omega_i = 1$. Households have access to financial markets and can smooth their consumption over time by holding or sellings assets a_t . The maximization problem is

$$\max_{\{c_t\}_{t \geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (1 - \zeta)^t \left(\frac{c_t^{1-\sigma} - 1}{1 - \sigma} + \gamma \frac{(\frac{a_t}{A_t} + \zeta)^{1-\Sigma}}{1 - \Sigma} \right),$$

subject to

$$c_t + a_{t+} = (1 - \tau_t)(z_t e_t N_t)^{1-\phi} + (1 + r_t^a + \text{cap}(a_{t-1}))a_{t-1} + \chi_t + \psi_t,$$

and the borrowing constraint $a_{t+1} \geq 0$. r_t^a is the *ex-post* return on wealth a_{t-1} at time t , c_t is the consumption choice, N_t is the labor supply, e_t is the idiosyncratic productivity type, χ_t is a lump-sum transfer from the government, ψ_t is an lump-sum inheritance financed by accidental bequest, and $\text{cap}(a_t)$ is the income received from unexpected capital gains, following an exogenous rule, defined later.

A key assumption in this model is that households derive utility from their relative wealth position a_t / A_t . This taste for wealth is non-homothetic, since we allow the coefficient on wealth Σ to be different from the coefficient on consumption σ , and typically assume $\sigma > \Sigma$. The Stone-Geary shifter ζ also creates some non-homotheticity, by decreasing the marginal utility of wealth for poorer households. We explore the implications of this assumption further in Section 3 and Section 4.

Due to the death-probability, a share ξ of all households die every period, leaving accident bequest, and are replaced by new-born households with zero wealth. We assume accidental bequests are transfered lump-sum to surviving households. This mortality risk is not essential for any of our results, but allow us to keep the wealth distribution non-degenerate while allowing rich households to keep accumulating wealth because of the non-homothetic taste for wealth.

In this framework, an increase in permanent labor income inequality takes the form of a mean-preserving increase in the spread of $\{z_i\}$, satisfying $\sum_i z_i \omega_i = 1$. The idiosyncratic productivity shock e_t follows an AR-(1) process that we discretize using the Tauchen method, with $\mathbb{E}[e_t] = 1$.

Firms. The supply side is composed of two sectors, a final goods sector and an intermediate goods sector. The final goods producer is in perfect competition, and produces using a continuum of imperfectly substitutable inputs j coming from the intermediate goods

sector

$$Y_t = \left(\int_j y_{jt}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

Firms in the intermediate goods sector produce using labor $y_{jt} = l_{jt}$. Assuming a symmetric equilibrium, we have $y_{jt} = y_t = Y_t$, and $l_{jt} = l_t = L_t$. Assuming flexible prices, this Dixit-Stiglitz framework implies that dividends are a constant share of output:

$$d_t = Y_t - w = Y_t \frac{\mu - 1}{\mu},$$

where $\mu = \frac{\varepsilon}{\varepsilon-1}$ is the markup, and the (constant) real wage is $w = \frac{1}{\mu}$. The value of the firm p_t is given by the non-arbitrage condition between equity and bonds

$$\frac{p_{t+1} + d_{t+1}}{p_t} = 1 + r_t.$$

Central bank. The monetary authority follows a real rate rule keeping the interest rate constant

$$r_t = r_{ss} + \varepsilon_t^r,$$

where ε_t^r is an exogenous policy shock. This is equivalent to assuming that the central bank controls inflation perfectly, or a limit case of a Taylor rule $i_t = r + \phi\pi_t$ where $\phi \rightarrow 1$. In both cases, this rule makes inflation irrelevant for all real variables in the model (Auclert et al. 2024).

Government. The fiscal authority sets a tax rate τ on labor income $L_t w$ and dividends d_t , emits real bonds B_t , spends public spendings G , and transfers χ_t . It also redistributes labor income according to a HSV tax function (Heathcote et al. 2017), so that total tax receipts are¹

$$T_t = \tau d_t + L_t w - (1 - \tau) \int (L_t w z_i e_t)^{1-\phi} d\mu_t.$$

¹One way to understand this equation is that the government fully taxes labor income $L_t w$, and then redistributes an amount $(1 - \tau)(w L_t z_i e_t)^{1-\phi}$ to households depending on their idiosyncratic and permanent productivity type.

The budget constraint writes

$$B_t = B_{t-1}(1 + r_{t-1}) + \chi_t + G - T_t.$$

Note that both the tax rate and government spendings are constant over time, but that tax receipts are still time-dependent due to the potential changes in dividends and labor income. We also assume two different fiscal rules. When facing a monetary policy shock, the government follows a balanced budget rule, so that $B_t = B_{t-1} = B_{ss}$, and only adjust transfers $\chi_t = T_t - B_{ss}r_{t-1} - G$. We follow Angeletos et al. (2024) and assume that, during a fiscal policy shock, the government adjusts the level of public debt B_t and transfers χ_t according to the following fiscal rule

$$\chi_t = \chi_{ss} + \tau(Y_t - Y_{ss}) - \varphi(B_{t-1} - B_{ss}) + \varepsilon_t^\chi,$$

with $\chi_{ss} = 0$ at the steady-state and ε_t^χ is an exogenous policy shock.

Mutual funds. A mutual fund receives all savings from households a_t , and invest them in the two assets available in the economy: government bonds B_t and firms equity p_t , such that $A_t \equiv \int a_{it}(a, e) d\mu_{it}(a, e) = p_t + B_t$. Assuming perfect competition, the funds make zero profits and financial income paid on savings to households $(1 + r_t^a)$ will be equal to bonds payments and dividends received from the firms and the government:

$$(1 + r_t^a)A_t = p_t + d_t + (1 + r_{t-1})B_t.$$

However, following an unexpected shock, the value of the firm in period 0 will jump, creating capital gains, and we will thus have

$$A_{ss}(1 + r_0^a) = p_0 + d_0 + (1 + r_{ss})B_{ss}.$$

The unexpected capital gains $p_0 + d_0 - p_{ss} - d_{ss}$ are paid to households according to a sigmoid function

$$cap_0(a_-) = \frac{K_0}{1 + \exp(-K_1(\log(a_-) - -K_2))},$$

where we impose that $\int cap_0(a_-)a_- = dp_0 + dd_0$. This rule allows us to parsimoniously capture the fact that richer households tend to invest a larger share of their wealth into equity, and thus receive a larger share of capital gains.

Unions. We assume that nominal wages are perfectly rigid, and a union sets a common labor supply for households in the economy. Due to the wage rigidity, the firm is demand-constrained and any variation in effective demand $C_t + G$ generates a one-to-one relationship with the labor demand N_t . This wage rigidity thus translates into a complete price rigidity.

Market clearing. We have three markets in this economy:

1. A goods market, satisfying $Y_t = C_t + G$,
2. An asset market, satisfying $A_t = B_t + p_t$,
3. A labor market, satisfying $L_t = N_t$.

We define the measure $\mu_{i,t}(a, e)$ as the mass of household of permanent type i holding wealth a with idiosyncratic type e at time t and $c_{i,t}(a, e), a_{i,t}(a, e)$ as the policy functions for consumption and savings for an agent holding wealth a with idiosyncratic type e at time t . Aggregate consumption and savings are then given by:

$$C_t = \sum_i \omega_i \int c_{i,t}(a, e) d\mu_{i,t}(a, e),$$

$$A_t = \sum_i \omega_i \int a_{i,t}(a, e) d\mu_{i,t}(a, e).$$

Definition 1. A competitive equilibrium is defined as policy functions for consumption $\{c_{i,t}(a, e)\}_{i,t}$ and savings $\{a_{i,t+1}(a, e)\}_{i,t}$; sequences for labor demand $\{L_t\}$, the real interest rates $\{r_t, r_t^a, cap_0\}$, the linear tax rate $\{\tau_t\}$, and output $\{Y_t\}$; a measure $\{\mu_{i,t}(a, e)\}_{i,t}$ such that:

1. Households solve their problem given prices and labor demand;
2. The representative firm maximizes profits;
3. The mutual funds balance sheet holds;

4. *The government budget constraint holds;*

5. *Markets clear:*

(a) *The asset market clears: $A_t = B_t + p_t, \quad \forall t$*

(b) *The goods market clears: $Y_t = C_t + G, \quad \forall t$*

(c) *The labor market clears: $L_t = N_t = 1 \quad \forall t$*

2 The Intertemporal Keynesian Cross and Permanent Labor Income Inequality

The output response of this model following a monetary policy shock can be described by an intertemporal Keynesian cross as in Auclert & Rognlie (2020). Indeed, note that aggregate consumption at time t can be written as

$$C_t \equiv \mathcal{C}_t(\{r_t^a, N_t, \chi_t, cap_0\}_{s=0}^\infty)$$

where $\mathcal{C}_t(\{r_t^a, N_t, \chi_t, cap_0\}_{s=0}^\infty)$ is the aggregate consumption policy function that depends on all the future paths of the real interest rate, labor income, transfers, and initial capital gains. We assume that the function $\mathcal{C}_t : \ell^\infty \times \ell^\infty \rightarrow \ell^\infty$ is Fréchet-differentiable. Totally differentiating the aggregate consumption function, we obtain

$$d\mathcal{C}_t(\{r_t^a, N_t, \chi_t, p_0\}_{s=0}^\infty) = \sum_{s=0}^\infty \frac{\partial \mathcal{C}_t}{\partial r_s^a} dr_s^a + \sum_{s=0}^\infty \frac{\partial \mathcal{C}_t}{\partial N_s} dN_s + \sum_{s=0}^\infty \frac{\partial \mathcal{C}_t}{\partial \chi_s} d\chi_s + \frac{\partial \mathcal{C}_t}{\partial cap_0} dcap_0.$$

Stacking it for all t , we obtain

$$d\mathbf{C} = \mathbf{M}^r d\mathbf{r} + \mathbf{M}^N d\mathbf{N} + \mathbf{M}^\chi d\boldsymbol{\chi} + \mathbf{m}^{cap} dcap_0,$$

where $\mathbf{M}^r$, $\mathbf{M}^N$ and $\mathbf{M}^\chi$ are aggregate sequence-space Jacobians (infinite-dimensional matrices) which characterize the reaction of aggregate consumption facing shocks on the real

interest rate and on net labor demand at different periods, as in Auclert et al. (2021, 2024):

$$\mathbf{M}^x \equiv \begin{pmatrix} \frac{\partial \mathcal{C}_0}{\partial x_0} & \frac{\partial \mathcal{C}_0}{\partial x_1} & \dots \\ \frac{\partial \mathcal{C}_1}{\partial x_0} & \frac{\partial \mathcal{C}_1}{\partial x_1} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}, \quad \mathbf{m}^{cap} \equiv \begin{pmatrix} \frac{\partial \mathcal{C}_0}{\partial cap_0} \\ \frac{\partial \mathcal{C}_1}{\partial cap_0} \\ \vdots \end{pmatrix}.$$

Note that we can also define the aggregate consumption function as the sum of the consumption functions for each permanent type *by unit of permanent income* $\mathcal{C}_t \equiv \sum_i \omega_i z_i \mathcal{C}_{i,t}$. It follows that the aggregate-level sequence space jacobian can be written as the following weighted average: $\mathbf{M}^x = \sum_i \omega_i z_i \mathbf{M}_i^x$. $\forall x \in \{r, L, \chi\}$, so that

$$d\mathbf{C} = \sum_i \omega_i z_i \left[\mathbf{M}_i^r d\mathbf{r} + \mathbf{M}_i^n d\mathbf{N} + \mathbf{M}_i^\chi d\chi + \mathbf{m}_i^{cap} dcap_0 \right].$$

Proposition 1. *An equilibrium following a monetary shock $d\mathbf{r}$ or a fiscal shock $d\chi$ is a sequence $d\mathbf{Y}$ such that*

$$d\mathbf{Y} = d\mathbf{C}$$

with $d\mathbf{C}$ satisfying Equation 2 and $d\mathbf{N} = d\mathbf{Y}$. In the case of a fiscal policy shock, $d\mathbf{r} = 0$ and $d\chi$ satisfies the budget constraint of the government. In the case of a monetary policy shock, $d\chi = d\mathbf{T} - Bd\mathbf{r}$.

Proof. Appendix A.1 □

Proposition 1 defines the output response in our HANK model as a function of the sequence space Jacobians. We can recover from this expression the traditional channels of both monetary and fiscal policy. $\mathbf{M}^r$ captures the direct effect of monetary policy, the change in consumption plans induced by a change in the real interest rate, keeping N_t and χ_t constant. A change in the path of the real interest rate has two direct effects: an intertemporal substitution effect and an income effect. By changing relative prices of consuming at different time periods, the new path of the real interest rate induces a change in consumption today. The change in the path of the real interest rate also changes households' financial income, implying a further change in consumption (income effect).

A change in the path of $\mathbf{r}$ also affects the budget constraint of the government, which will imply a direct change in the path of transfers χ , further changing the consumption path. The total change in the consumption path of households changes aggregate demand, which changes the firm's labor demand and hence the labor income of all households. This second change in consumption, in turn, has a feedback effect on aggregate demand, which is the traditional Keynesian multiplier effect. This general equilibrium amplification is the indirect effect from labor demand, and is captured in the IKC by the matrix $\mathbf{M}^N$. Finally, a change in output implies a change in profits and the valuation of the firms, creating capital gains at $t = 0$, captured by the vector $\mathbf{m}^{cap}$.

For fiscal policy, a change in the path of $\mathbf{M}^\chi$ increases income today and decreases income in the future, when the government taxes households through negative transfers. We assume that the intertemporal budget constraint of the government holds and that all deficits are financed by future tax receipts: $\sum (1+r)^t (d\chi_t - dT_t) = 0$. As some households are constrained in our model, this change in the path of income increases consumption, so that $\mathbf{M}^\chi d\chi$ is positive. This change in consumption, in turns, increases output, and hence the labor income of households, which further increases consumption and output.

Using the IKC as an ordering device, we ask: what happens when permanent labor income inequality goes up? We compare the output response following a monetary and fiscal shock of two economies, one characterized by a low level of permanent labor income inequality $\mathbf{z}$, and one characterized by a higher level of permanent labor income inequality $\mathbf{z}'$. A bold $\mathbf{z}$ indexes economies with different levels of permanent labor income inequality. Proposition 2 shows that we can decompose this change in the output response to a fiscal and monetary shock between three effects: (1.) a composition effect, (2.) a Jacobian effect and (3.) an amplification effect.

Proposition 2. *The difference in the output response following a monetary policy shock in two economies with different levels of permanent income inequality can be decomposed between three terms: (1.) a direct and an indirect composition effect, (2.) a direct and an indirect Jacobian effect,*

and (3.) an amplification effect:

$$\begin{aligned}
dY(\mathbf{z}') - dY(\mathbf{z}) = & \underbrace{\sum_i \omega_i \Delta z'_i \left[\overbrace{\mathbf{M}'_i(\mathbf{z}') d\mathbf{r}}^{\text{direct}} + \overbrace{\mathbf{M}^N_i(\mathbf{z}') d\mathbf{N}(\mathbf{z}') + \mathbf{M}^X_i(\mathbf{z}') d\chi + \mathbf{m}^{cap}_i dcap_0}^{\text{indirect}} \right]}_{\text{composition effect}} \\
& + \underbrace{\sum_i \omega_i z_i \left[\overbrace{\Delta \mathbf{M}'_i(\mathbf{z}') d\mathbf{r}}^{\text{direct}} + \overbrace{\Delta \mathbf{M}^N_i(\mathbf{z}') d\mathbf{N}(\mathbf{z}') + \Delta \mathbf{M}^X_i(\mathbf{z}') d\chi + \Delta \mathbf{m}^{cap}_i dcap_0}^{\text{indirect}} \right]}_{\text{Jacobian effect}} \\
& + \underbrace{\mathbf{M}^N(\mathbf{z}) \left(d\mathbf{N}(\mathbf{z}') - d\mathbf{N}(\mathbf{z}) \right)}_{\text{amplification effect}}.
\end{aligned}$$

With $\Delta \mathbf{M}^x(\mathbf{z}') = \mathbf{M}^x(\mathbf{z}') - \mathbf{M}^x(\mathbf{z})$ and $\Delta z'_i = z'_i - z_i$.

Proof. Appendix A.2 □

The composition effect captures the fact that even if the response of consumption *per unit of permanent income* for each type does not change, the increase in inequality will put more weight on the response of the high type compared to the low type.

The Jacobian effect captures the change in the coefficients of our sequences-space Jacobians keeping the weights on each matrices constant. The change in coefficients comes from three factors: (1) both types change their consumption-saving decisions as their income increases, (2) in general equilibrium prices adjust, and this change in prices also affects the consumption-saving decisions of households, and (3) the distribution of households also changes across the two economies.

More formally, the change in the type- i Jacobian's now depends on the change in the policy function *and* in the distribution of wealth $\mu_{i,t}$. Indeed, the definition of $M^x_{i,t,j}$, the element in line t , column j of the sequence-space Jacobian M^x_i , can be written as²

$$M^x_{i,t,j}(\mathbf{z}) = \int \frac{\partial c_{i,t}(a, e; \mathbf{z})}{\partial x_j} d\mu_{i,t}(a, e; \mathbf{z}).$$

²Note that the path of distribution $d\mu_t(a, e; s)$ itself depends on the derivative of the policy functions.

Table 1: Calibration of the HANK model

Parameter	Description	Value	Source
<i>Households</i>			
β	Discount factor	0.96	Annual r of 2%.
σ	CRRA on consumption	1.00	Standard.
Σ	CRRA on wealth	0.45	Pareto coeff. on cons./wealth.
γ	Strength of taste for wealth	0.79	Top 1%.
ζ	Stone-Geary shifter	0.23	Aggregate MPC.
ξ	Death probability	0.01	Pareto coeff. on cons./wealth.
ρ^e, σ^e	Corr. and var. of shocks (annual)	(0.91, 0.5)	
$\{K_0, K_1, K_2\}$	Portfolio shares	(0.66, 0.79, 1.87)	SCF (2019).
<i>Government and Firms</i>			
B_{ss}	Public debt	4.00	B/Y 100%.
G_{ss}	Gov. Spendings	0.18	National accounts.
τ_{ss}	Tax rate	0.17	Residual.
ϕ	Tax progressivity	0.18	Standard.
φ	Fiscal rule parameter	0.03	Standard.
μ	Markup	1.09	Annual W/Y 4.47

Using this definition of a sequence-space Jacobian we can compute ΔM_i^x , the change in the Jacobian of an agent of type i in period t following a shock on the aggregate variable x in period j as

$$\begin{aligned}
\Delta M_{i,t,j}^x(\mathbf{z}') = & \underbrace{\int \left(\frac{\partial c_{i,t}(a, e; \mathbf{z}')}{\partial x_j} - \frac{\partial c_{i,t}(a, e; \mathbf{z})}{\partial x_j} \right) d\mu_{i,t}(a, e; \mathbf{z})}_{\text{policy-function effect (1)+(2)}} \\
& + \underbrace{\int \frac{\partial c_{i,t}(a, e; \mathbf{z})}{\partial x_j} \left(d\mu_{i,t}(a, e; \mathbf{z}') - d\mu_{i,t}(a, e; \mathbf{z}) \right)}_{\text{wealth-distribution effect (3)}}
\end{aligned}$$

Finally, in general equilibrium, the amplification effect magnifies the first two effects. It acts a multiplier of the sum of the composition and the Jacobian effect. Thus, even small Jacobian and composition effect might end up having a large impact on the output response, since they will be amplified by the general equilibrium effect.

3 Empirical Fit of the Model

The following section describes how we calibrate the model to match three empirical moments: (1) the concentration of wealth, (2) the aggregate wealth-to-output ratio, and (3) an annual marginal propensity to consume (MPC) of 0.5.

3.1 Calibration

We calibrate the model at quarterly frequency using U.S. data for 2019. Table 1 reports all parameter values.

Taste for wealth and death probability. The parameters governing the taste for wealth, γ , ζ , Σ , together with the death probability ξ , are set to match four targets: the observed Pareto tail ζ^a , the ratio of the consumption and wealth tails ζ^c / ζ^a , the top 10% wealth share of (71%), and an annual MPC of 0.5. Following Gaillard et al. (2023), we derive an analytical approximation of the Pareto tails in the model, which yields

$$\zeta^a = \frac{-\log(1 - \xi)}{\log(1 + r)}, \quad \zeta^c = \zeta^a \frac{\sigma}{\Sigma}.$$

Appendix B provides the derivation. Setting $\sigma = 1$, targetting $\zeta^a = 1.23$ and $\zeta^c = 2.24$, and using an annual interest rate r of 2%, we obtain $\Sigma = 0.445$ and $\xi = 0.006^3$. We then set the discount factor β so that the asset market clears.

Figure 1a shows that our approximation accurately captures the model's implied tail behavior: the model provides a good fit to the observed Pareto tail of wealth in the data and generates a significant amount of households with a nominal wealth level above a billion USD. Table 2 compares the model distribution with PSZ (2019). The calibration matches the targeted top 10% wealth share, and the low amount of wealth held by the bottom of the distribution. The calibration matches the top 10% wealth share and the low wealth holdings of the bottom half of the population, and also fits the top 0.1% and 0.01% shares reasonably well. It overstates the top 1% share, a point we return to below.

³This implies an annual death probability of roughly 2.5%, and thus an expected lifetime of 42 years.

Portfolio shares. Wealthy households hold a larger fraction of their assets in equity and business wealth, whose valuation increases when the risk-free interest rate declines. We follow Auclert & Rognlie (2020) and match this feature of the data by calibrating our exogenous portfolio share (see Equation 1) on the portfolio shares observed in the Survey of Consumer Finance. This yields the parameters $K_0 = 0.66$, $K_1 = 0.79$, $K_2 = 1.87$. Figure 1b compares the fitted function to the data. Note that in our main calibration, we rely on the narrow definition of capital.

Labor income inequality. Permanent labor income levels $\{z_i\}$ are set to match the labor income shares of the bottom 50%, the next 50%, the next 9%, and the top 10% reported in Piketty et al. (2018b). Idiosyncratic labor-income risk follows an AR(1) process with persistence $\rho_e = 0.95$ and innovation standard deviation $\sigma_e = 0.15$.

Other parameters. We normalize steady-state quarterly output to $Y_{ss} = 1$ and set public debt to $B_{ss} = 4$, consistent with a debt-to-output ratio of 100%. The markup μ is chosen to match an annual wealth-to-output ratio of 4.47.⁴ We set $\phi = 0.18$ and $G = 0.17$ following the literature, and determine the tax rate τ as the residual that closes the government budget constraint.

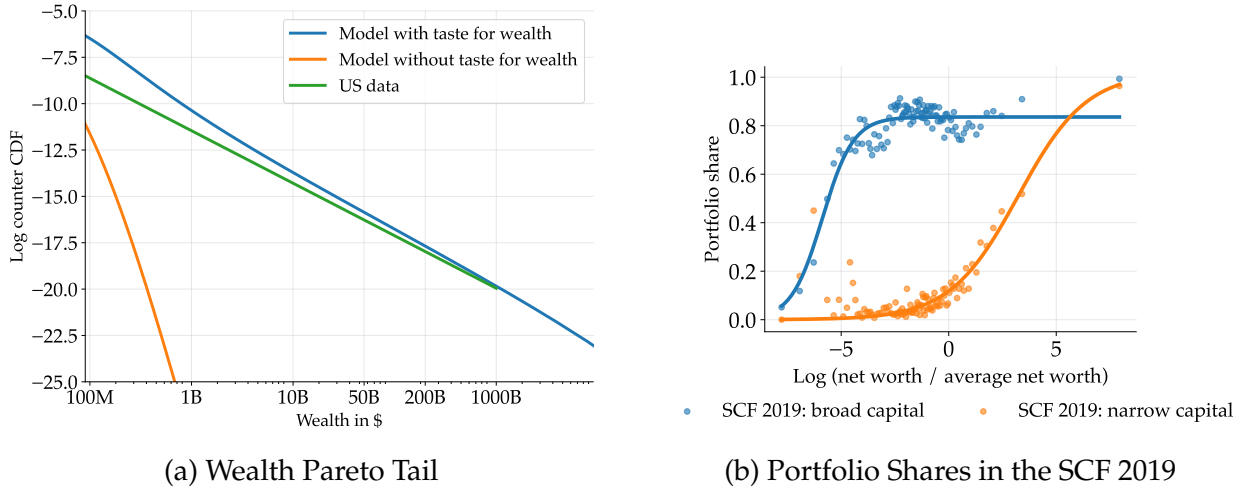


Figure 1: Wealth tail and portfolio composition.

Note: Panel (a) plots the Pareto tail obtained in the benchmark model, in the data (assuming $\zeta^a = 1.23$), and in a model with $\gamma = 0$. Panel (b) compares the calibrated sigmoid portfolio-share function to the portfolio shares computed in the Survey of Consumer Finances.

⁴The markup determines steady-state profits and therefore the equity value, $p_{ss} = d_{ss}/r_{ss}$.

3.2 The Taste for Wealth and the Wealth Distribution

It is well-known that standard HANK models generate too little wealth inequality and too low MPCs (Kaplan et al. 2018, Auclert et al. 2024). In these environments, idiosyncratic risk and precautionary saving are the only sources of heterogeneity: households accumulate buffers up to a target level of wealth and then dissave at a rate that prevents the emergence of very rich households.

Table 2: Wealth distribution for the year 2019

	Bottom 50%	Middle 40%	Top 10%	Top 1%	Top 0.1%	Top 0.01%	MPC	HtM
Data PSZ	0	28	71	35	17	9		
Model	0	28	72	44	22	10	0.5	0.48

These models thus cannot reproduce the high savings rates observed empirically at the top of the wealth distribution. When the calibration is adjusted to match the aggregate wealth-to-output ratio, too much wealth is assigned to the lower and middle parts of the distribution, mechanically lowering their MPCs. The failure to match high saving rates at the top and an high aggregate MPC are thus two sides of the same issue.

Introducing a taste for wealth resolves this tension by providing an additional saving motive that is strongly increasing in wealth. Because households derive direct utility from wealth, and because this motive is non-homothetic, the very rich behave as if wealth were a luxury good: once their consumption needs are met, they devote almost all marginal income to additional saving. This mechanism allows the model to match the high saving rates documented for the top of the distribution (Dynan et al. 2004, Kumhof et al. 2015, Straub 2019).

Figure 2 shows the policy function for next-period assets as a function of wealth and permanent income. As wealth increases, savings approach the simple accumulation rule $a_t \approx (1 + r)a_{t-1}$, indicating that wealthy households save nearly all of their disposable resources. The death probability then plays a crucial role in keeping the distribution stationary, as it prevents indefinite wealth accumulation from rich households ⁵.

⁵It can thus be interpreted as a “disaster” shock that resets wealth to zero, with the key difference that

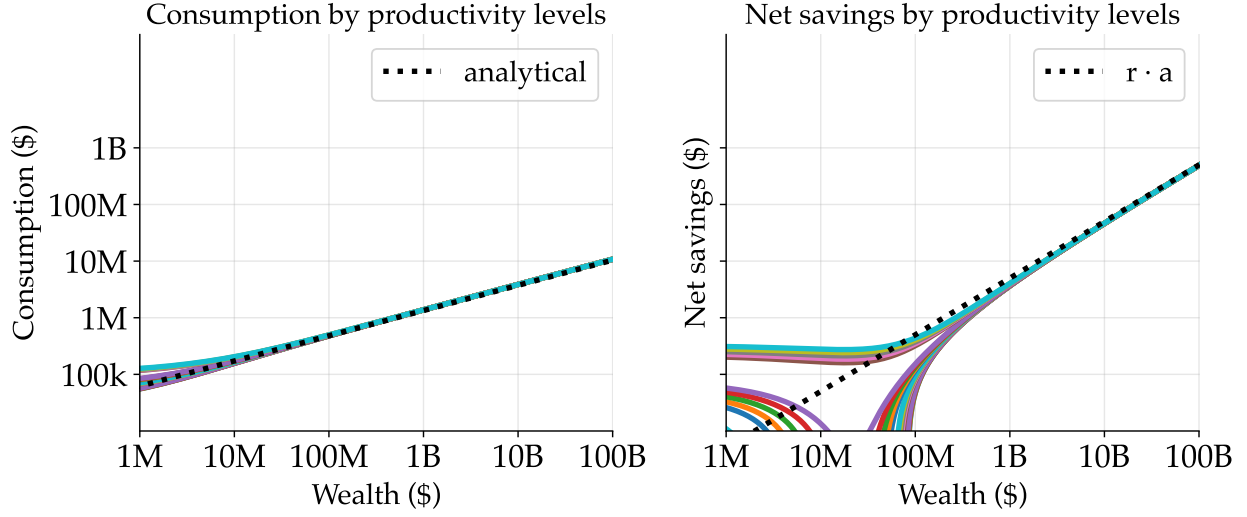


Figure 2: Savings and Consumption Behavior of Wealthy Households

Note: This figure plots the numerical policy functions for wealth and consumption in the model as a function of nominal wealth, expressed in dollars (x-axis), idiosyncratic and permanent types (different colors). Dashed black lines plot the approximation for those policy functions as $\text{wealth} \rightarrow \infty$, as shown in Appendix B.

Households at the bottom of the wealth distribution behave differently. Because the taste for wealth is negligible at low wealth levels, their decisions resemble those of standard buffer-stock agents. Their saving is driven primarily by precautionary motives. As the strong saving behavior of wealthy households pushes the equilibrium return to a level such that $\beta(1+r) \ll 1$, even with modest idiosyncratic risk, this substantially reduces the incentive for low-wealth households to save and keeps them near the borrowing constraint. In equilibrium, this produces a large mass of hand-to-mouth households and yields an aggregate MPC close to 0.5.

Finally, our calibration overstates the wealth share of the top 1%. This reflects a version of the "missing middle" problem commonly observed in one-asset models: it is difficult to simultaneously match the extreme right tail and the wealth holdings of upper-middle groups such as the top 10%, whose portfolios are dominated by housing. Because we target the top 10% share, the model necessarily allocates too much mass to the very top.

the associated marginal utility is zero rather than extremely large, as would be the case with a literal -100% return shock.

4 Impact of Rising Permanent Labor Income Inequality on Stabilization Policies

This section studies how a rise in permanent labor income (PLI) inequality changes the transmission and effectiveness of stabilization policies in our HANK model. The main experiment compares two economies that are identical along all dimensions except for the distribution of PLI: a “low-inequality” economy calibrated to the U.S. in 1970 and a “high-inequality” economy calibrated to the U.S. in 2019. In both cases we compute impulse responses to a monetary policy shock (a cut in the real interest rate) and to a fiscal shock (a temporary increase in transfers), and we decompose the mechanisms using the intertemporal Keynesian cross representation of the model.

4.1 Steady-State Changes

Rising permanent labor income (PLI) inequality alters the steady state of the economy in ways that directly affect the Jacobian components of the intertemporal Keynesian cross. The decomposition developed in Section 2 shows that the effect of a policy shock can be attributed to: (i) a composition effect, (ii) a Jacobian effect, and (iii) a generalequilibrium amplification effect. In what follows, we abstract from the composition and amplification components and focus on how PLI inequality modifies the Jacobians themselves—that is, the elements of $\mathbf{M}^r$, $\mathbf{M}^N$, and $\mathbf{M}^X$ that encode the marginal propensities to consume (MPC), the intertemporal elasticity of substitution (EIS), and the sensitivity of consumption to transfers.

Figure 3b reports the steady-state MPC and EIS for each permanent type, aggregated over wealth and idiosyncratic shocks and evaluated at constant prices. Two patterns emerge:

1. The MPC is a *decreasing concave function* of PLI.
2. The EIS is a *increasing concave function* of PLI.

By Jensen’s inequality, a mean-preserving increasing in the spread of PLI inequality therefore *raises* the aggregate MPC and *lowers* the aggregate EIS.

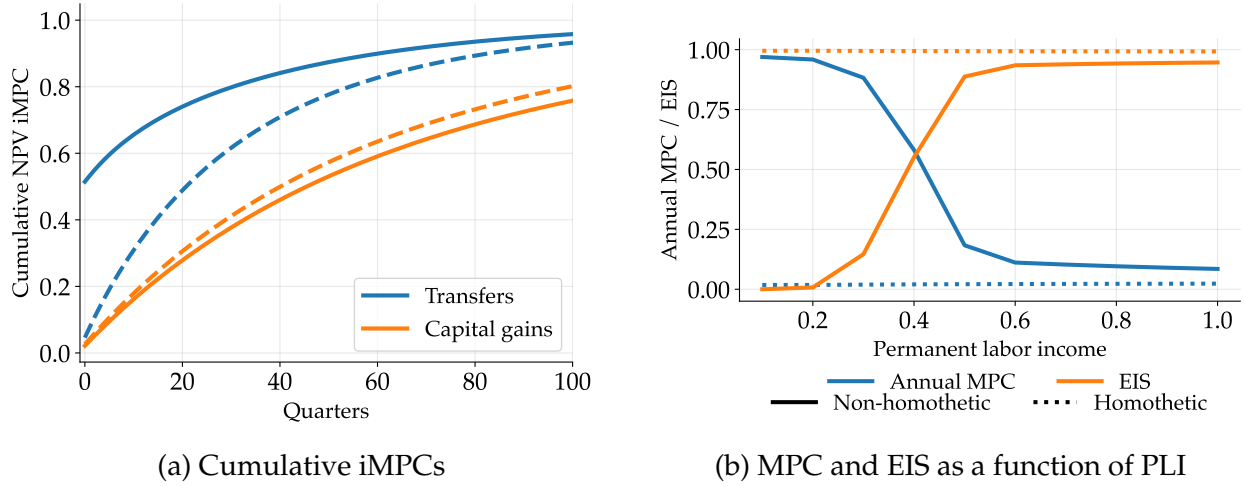


Figure 3: Change in MPC and EIS

Note: The left panel (a.) shows the cumulative net-present value iMPCs for transfers and capital gains, in the model with high-inequality (full line) and low-inequality (dashed-line). The right panel (b.) shows the EIS and the annual MPC as a function of permanent labor income in a model with non-homothetic preferences ($\gamma > 0$) and homothetic preferences ($\gamma = 0$). To compute those statistics, we keep prices constant, solve the households problem for different values of permanent labor income, and aggregate over the distribution wealth and idiosyncratic shocks.

These changes show up clearly in the steady-state statistics. As shown in Tables 3 and 4, at the permanent-type level, the EIS of the bottom 50% type falls from about 0.84 in the low-inequality calibration to about 0.68 in the high-inequality calibration, while the EIS of richer types remains close to one. Aggregating across types, the EIS declines from roughly 0.91 to 0.83. The opposite pattern holds for MPCs: the quarterly MPC of the bottom 50% type increases from about 0.13 to 0.27 when moving from low to high PLI inequality, and the aggregate MPC roughly doubles, from around 0.07 to 0.14.

The Jacobian effect itself can be decomposed into (i) changes in households' policy functions and (ii) changes in the steady-state wealth distribution. Figure 4 illustrates this decomposition by plotting the MPC and EIS as functions of both PLI and wealth, and comparing these objects to the shift in the wealth distribution across calibrations. Holding wealth fixed, increasing PLI alters MPCs and EIS only modestly. By contrast, both objects vary sharply with wealth. Because PLI inequality induces a large reallocation of wealth toward high-PLI households and compresses the wealth holdings of low-PLI households, the resulting wealth-distribution effect accounts for most of the movement in the Jacobians.

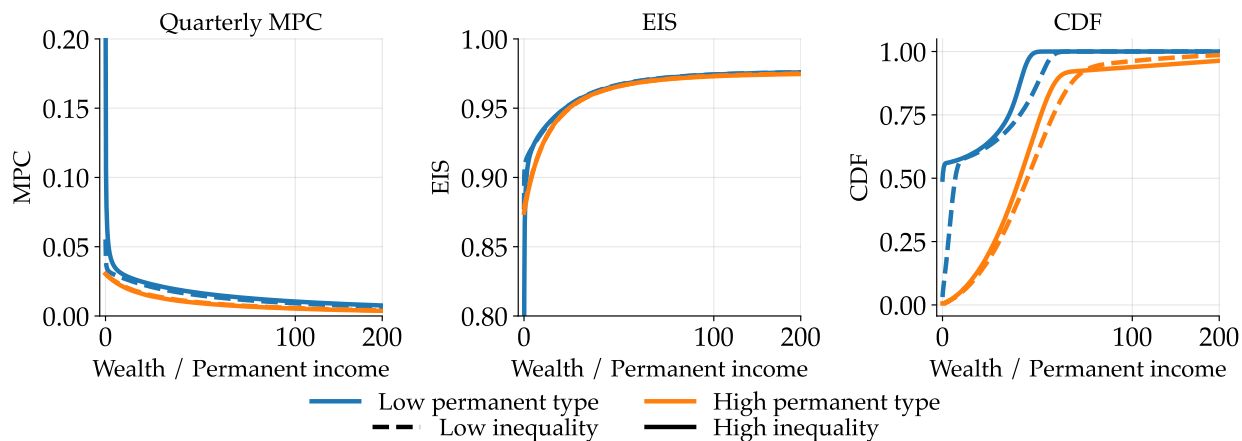


Figure 4: Net-present value cumulative iMPCs with low and high PLI inequality

Note: The left figure shows the quarterly MPC as a function of wealth (x-axis) and permanent income (blue and orange lines), in a case with high inequality (full lines) and low inequality (dashed-lines). The middle figure shows a similar information for the EIS. The last figure shows the cumulative density function for wealth for the low/high permanent labor income type, in both economies.

The mechanism is as follows. When PLI becomes more unequal, high-PLI households receive a larger share of aggregate labor income and—due to the non-homothetic taste for wealth—save a large fraction of these resources. Their additional saving pushes aggregate wealth up and depresses the equilibrium real interest rate. Low-PLI households now face both a lower permanent income and a lower return on saving, which reduces their target wealth, closer to the borrowing constraint. This generates a larger mass of hand-to-mouth households, higher average MPCs, and lower aggregate EIS. In short, rising PLI inequality shifts wealth and discounting behavior toward agents who are less forward-looking and more sensitive to income shocks. Consistent with this mechanism, Figure 3a shows that cumulative iMPCs become more front-loaded in the high-inequality calibration.

Together, these forces produce a steady state in which (i) aggregate consumption is more sensitive to current income and transfers, and (ii) substantially less responsive to interest rate shocks. These changes in the Jacobian structure are central for understanding the policy experiments that follow.

Table 3: Intertemporal Elasticity of Substitution (IES)

	Bottom 50 type	Next 40 type	Top 10 type	Top 1 type	Aggregate
Low inequality	0.91	0.95	0.96	0.96	0.93
High inequality	0.37	0.95	0.96	0.96	0.66

Table 4: Change in the MPC

	Bottom 50 type	Next 40 type	Top 10 type	Top 1 type	Aggregate
Low inequality	0.15	0.06	0.05	0.03	0.1
High inequality - PE	0.56	0.07	0.04	0.02	0.31
High inequality - GE	0.74	0.08	0.05	0.02	0.4

4.2 Monetary Policy

We examine the effect of a expansionary monetary policy shock which temporarily decreases the real interest rate. The shock follows an AR-1 process with initial impulse σ^r of 25 basis points and persistence $\rho^r = 0.9$. Using the intertemporal Keynesian cross representation, the consumption response to this shock can be decomposed into three main channels: the direct effect of the change in $d\mathbf{r}$, the indirect general equilibrium effect operating through labor income $d\mathbf{N}$ and government transfers $d\chi$, and the capital gains channel associated with asset prices and valuations. Figure 5a displays this decomposition for the low- and high-PLI economies.

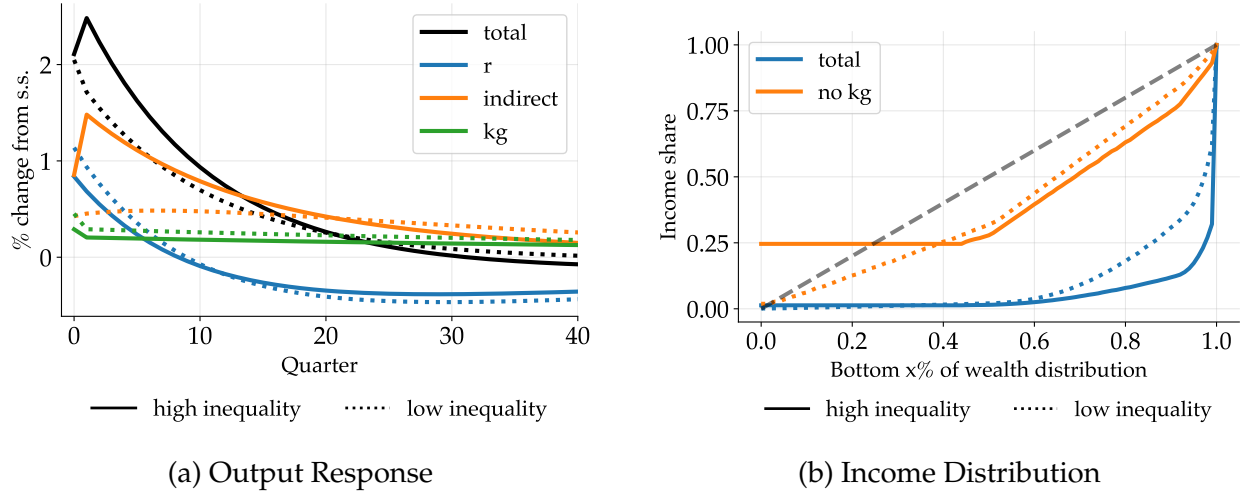


Figure 5: Output and Income Distribution Response to a Monetary Policy Shock

Note: The left panel (a.) shows the decomposition of the output response to a monetary policy of 25 b.p. with persistent 0.9. Full lines correspond to the high-inequality case, dashed lines correspond to the low-inequality case. The right panel (b.) shows the distribution of the income generated in the first period of the shock, dY_0 , as a function of wealth. The red line shows the distribution of income without capital gains, the blue line shows the distribution of total income generated, including capital gains.

Direct effect. The direct effect — the response of consumption due to the change in the real interest rate holding incomes constant — declines when PLI inequality rises. This is the mechanical consequence of the lower aggregate EIS: households are less inclined to reallocate consumption over time in response to interest rate movements. With more households near the borrowing constraint and a larger mass of agents whose consumption is insensitive to intertemporal price movements, the sensitivity of consumption to changes in interest-rate diminishes. Consequently, the partial-equilibrium term $\mathbf{M}^r d\mathbf{r}$ accounts for a smaller fraction of the overall response in the high-inequality economy.

Indirect effect. By contrast, the indirect effect through labor income becomes more important. Because aggregate MPCs are higher and a larger mass of households is close to the borrowing constraint, any change in income induced by the monetary shock feeds back more strongly into consumption demand. An expansionary monetary shock increases output and labor income; with higher MPCs, these income losses translate into larger consumption increases, which further expands demand. This amplification raises the general equilibrium multiplier associated with monetary policy. In the decomposition, the contribution of the labor-income channel to the consumption response increases with PLI

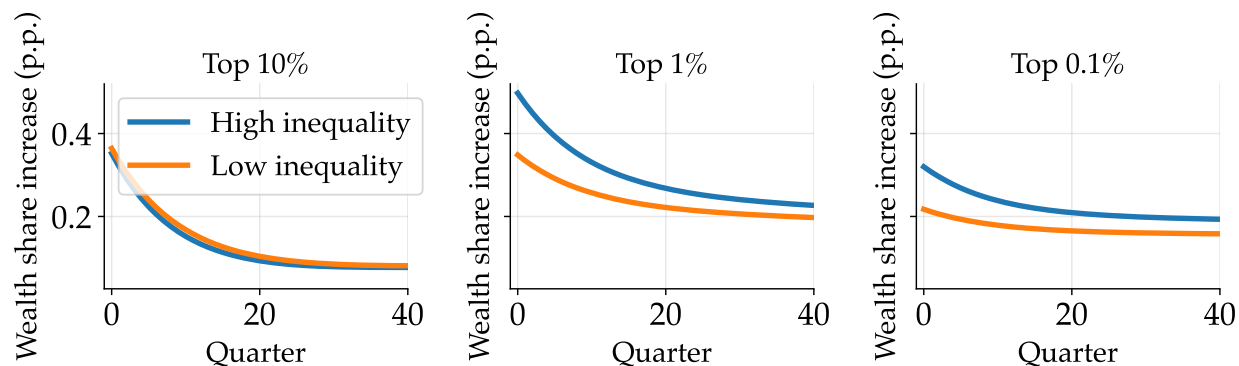


Figure 6: Distribution of Wealth following a Monetary Policy Shock

inequality.

Capital gains. The capital gains channel is relatively weaker for aggregate consumption but more powerful for the distribution of wealth. A monetary easing boosts asset prices and generates capital gains, which are disproportionately received by wealthy households with low MPCs. When PLI inequality is high, these households own a larger share of total wealth. As a result, the same asset price movement redistributes more towards the top and contributes less to aggregate consumption. In the decomposition, the capital gains channel makes a smaller positive contribution to aggregate spending in the high-inequality economy than in the low-inequality one, even though the underlying valuation effects are larger.

Aggregate effect. Overall, the increase in the indirect effect is larger than the decrease in the direct and capital gains effects, so that the overall output response to a given monetary shock increases. This is consistent with empirical findings Almgren et al. (2022) who find that countries in the euro-zone with a higher share of hand-to-mouth households also tend to a higher consumption response to monetary policy shocks.

Redistributional effects. The redistributive consequences of monetary policy also become stronger when PLI inequality is high. As shown in Figure 5b, the income generated by a monetary easing in the first period is disproportionately received by high-wealth households. This reflects the fact that asset-price movements are the dominant channel through which monetary policy redistributes resources, and these gains scale with wealth. Conse-

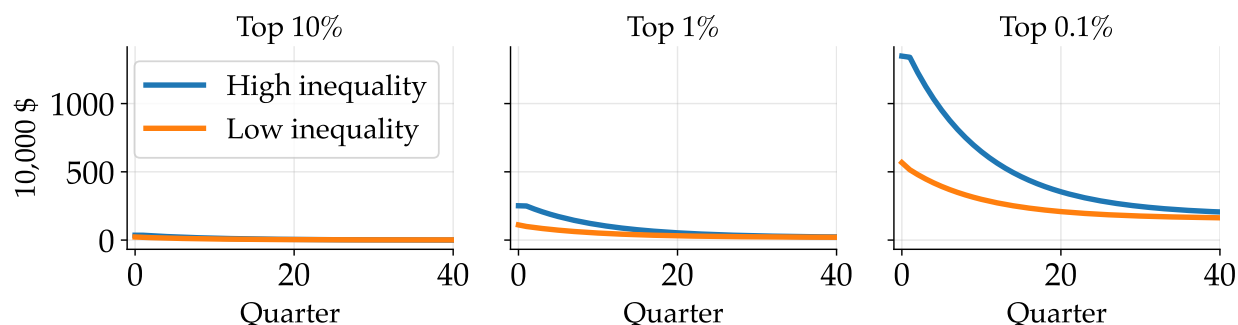


Figure 7: Wealth Levels following a Monetary Policy Shock

Note: This figure shows the increase in the average wealth levels of a household within the top 10%, top 1%, and top 0.1%, following a monetary policy shock, in USD.

quently, following an accommodative shock, a larger fraction of valuation gains accrues to households at the top of the distribution. The top wealth shares and the absolute wealth levels of rich households therefore respond more strongly in the high-inequality calibration (Figures 6 and 7). The model thus predicts that rising PLI inequality simultaneously amplifies the macroeconomic impact of monetary policy and increases its propensity to widen wealth inequality—consistent with empirical evidence that monetary easing operates in part through asset-price channels that disproportionately benefit the wealthiest households.

Note that the magnitude of this redistributive effect is economically relevant. Although our monetary policy shock is small (25 b.p.), the top 0.1% wealth share increases by 0.32 p.p. on impact. This represents an increase in wealth of 14 million USD for an average household within the top 0.1% in the model.

The model thus delivers two main implications. First, monetary policy becomes more powerful in the aggregate when PLI inequality is high, because larger MPCs amplify income-based feedbacks. Second, monetary easing becomes more redistributive, since valuation effects scale with wealth and the wealth distribution is more unequal. Both implications are consistent with empirical evidence that the asset-price channel of monetary policy disproportionately benefits households at the top of the wealth distribution.

4.3 Fiscal Policy

We next study the response to a fiscal shock, modeled as a temporary increase in transfers financed by future fiscal adjustments via the government's feedback rule (Equation 1). We focus on two properties of fiscal deficits: (i) the size of the cumulative multiplier, and (ii) the degree of self-financing.

Impact and cumulative multipliers. We first focus on a fiscal shock $d\tilde{\chi}_t = \rho_\chi^t d\chi_0$, where $\tilde{\chi}$ is a shock to transfers, and assume that total transfers adjust so that $dB_t = \rho_B(dB_{t-1} + d\tilde{\chi}_t)$. If ρ_B is high, the deficit is very persistent and the increase in negative transfers to pay back the debt is postponed further in time. If ρ_B is low, the deficit is short, and total transfers $d\chi_t$ adjust quickly to bring back the debt to steady-state. We define the impact multiplier as $dY_0/d\tilde{\chi}_0$ and the cumulative multiplier as $(\sum_t Q^t dY_t)/(\sum_t Q^t d\tilde{\chi}_t)$ with $Q = 1/(1+r)$.

Figure 8 shows the impact and cumulative multiplier, in the low inequality and high inequality economies, as a function of $d\rho_B$. As shown previously by Auclert et al. (2024), the cumulative multiplier is an increasing function of ρ_B . The classical mechanism behind this result is that, due to the borrowing constraint, Euler equation is more "discounted" in the terms of McKay et al. (2017) and Michaillat & Saez (2021). Future negative transfers are discounted more than the short-term increase in income due to the deficits-financed transfers, which increases output in the short term, which, in turns, increases income and consumption further.

When permanent labor income inequality increases, the share of hand-to-mouth households increases and the likelihood to become constraint increases for low PLI households. They discount the future by even more, and the iMPC becomes more front-loaded, as shown in Figure 3a. This magnifies the effect of persistent deficits even further, so that cumulative multipliers increase by more with ρ_B in the high inequality case compared to the low-inequality case.

Self-financing deficits. We now investigate the possibility of self-financing deficits in our model. We consider the definition of self-financing deficits proposed by Angeletos et al.

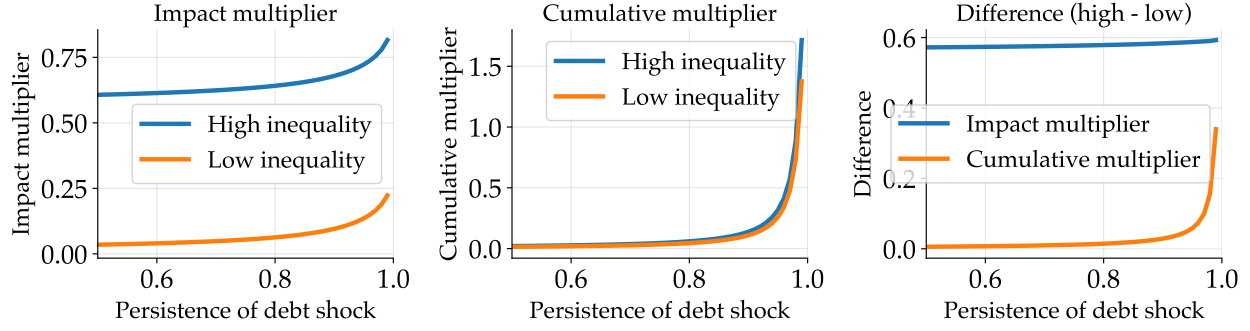


Figure 8: Impact and Cumulative Multipliers

Note: The left panel shows the impact multiplier $dY_0/d\chi_0$, the middle panel shows the cumulative multiplier $(\sum_t Q^t dY_t)/(\sum_t Q^t d\chi_t)$ with $Q = 1/(1+r)$. The right panel shows the difference between the high and low inequality case.

(2024):

$$\nu \equiv \frac{\sum_{t=0}^{\infty} (1+r)^{-t} dT_t}{\varepsilon_0^\chi}.$$

That is, we compute the ratio of the net-present value of tax receipts induced by a small deficit financed transfers ε_0^χ in period 0. Note that, due to the real short-term bonds assumption in our model, there is not debt-erosion channel due to higher inflation. We assume that, following this shock, the government follows the fiscal rule defined by Equation 1.

The main intuition behind the self-financing mechanism is that a small deficit in period 0 increases output in the short term. Even if the government keeps the tax rate τ constant, this increase in output will increase tax receipts, which are in our model, approximately proportional to τdY_t . If this increase in tax receipts is large enough, the government might fully finance the deficits through higher income receipts coming from the increase in the tax base, instead of having to adjust transfers negatively in the future (which correspond to a case where $\nu = 1$).

Figure 9 plots the degree of self-financing as a function of φ , the parameter in our fiscal rule governing how quickly transfers adjust when public debt is above its steady-state level. When φ is high, transfers become negative quickly when public debt is high, so that deficits are short-lived. When φ is low, transfers adjust further in the future, and

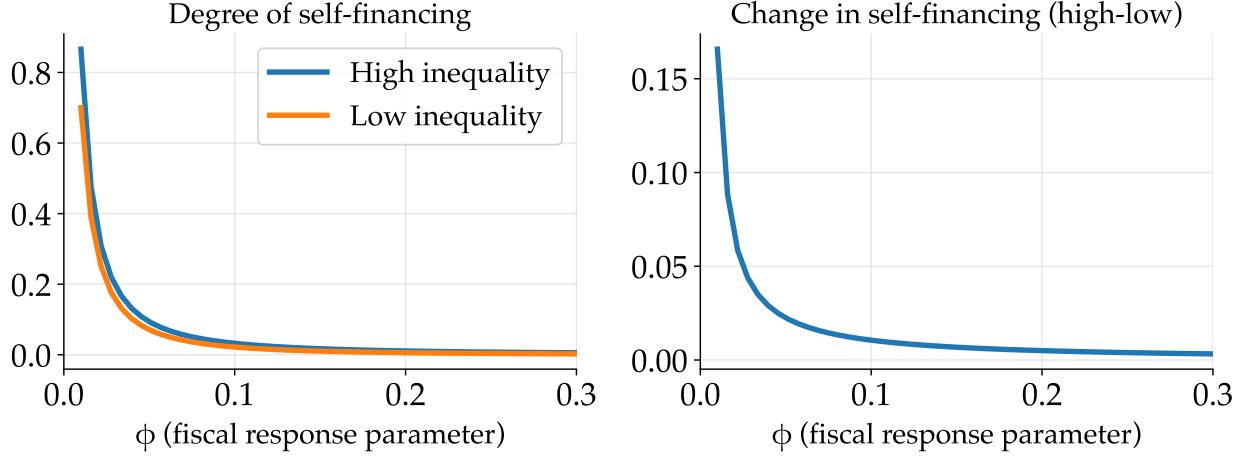


Figure 9: Self-financing deficits

Note: The left panel shows the degree of self-financing ν in the low and high-inequality case, as a function of ϕ . The right panel plots the difference between those two curves.

public debt remains higher than its steady-state level for longer.

As in Angeletos et al. (2024), the degree of self-financing decreases with ϕ . This is due to the same property highlighted in the previous section studying cumulative multipliers: the iMPCs in HANK models are naturally front-loaded, so that future negative transfers are discount by more than current positive transfers. An increase in permanent labor income inequality makes iMPCs more front-loaded, and hence increase the likelihood of self-financing deficits.

4.4 Interaction between fiscal and monetary policy

Finally, we analyze how fiscal and monetary policy interact in the presence of rising PLI inequality. When the central bank decreases the real interest rate, it creates some fiscal space for the government by (i.) decreasing its cost of repaying its debt and (ii) increasing its tax receipts, through the endogenous increase in output. Until now, we have assume that, facing a monetary policy shock, the government kept the public debt constant, $dB_t = 0$, which can be seen as a particular case of Equation 1 with $\phi \rightarrow \infty$. In this section, we relax this assumption and study how the output response to a monetary policy shock adjusts when ϕ changes.

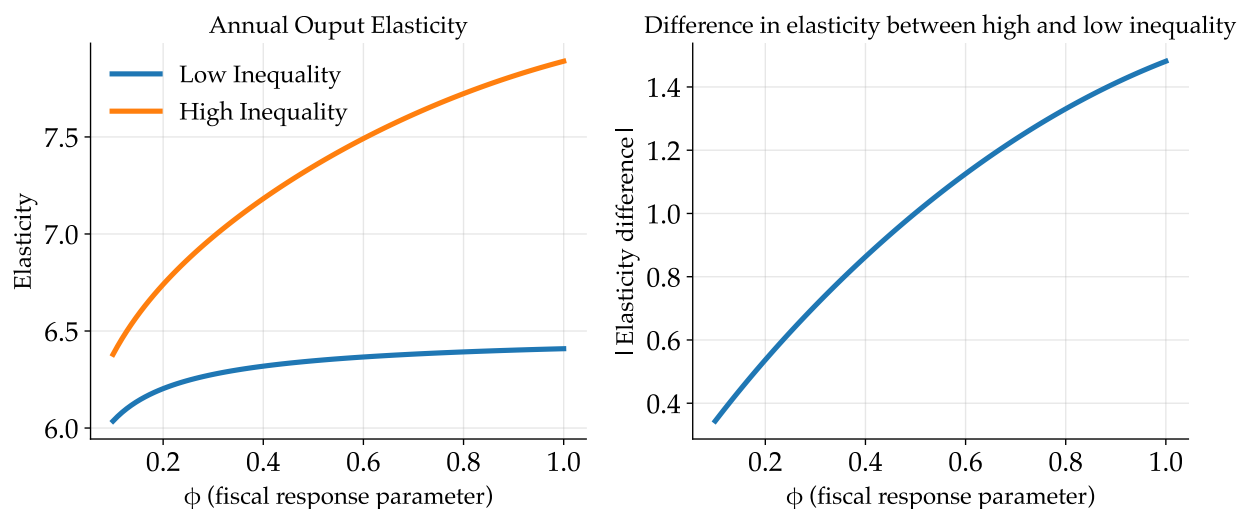


Figure 10: Change in Annual Output Elasticity to Monetary Policy Shock

Figure 10 shows the annual output elasticity to a monetary policy shock as a function of ϕ . When In contrast, when PLI inequality is high, fiscal responses to monetary shocks become central for determining the overall output effect. Because households are more hand-to-mouth and MPCs are higher, any adjustment in transfers has a large impact on aggregate demand. If fiscal policy accommodates monetary policy—for instance, by allowing transfers to remain relatively high in the aftermath of a rate cut—then the expansionary effect of monetary policy is strongly amplified. Conversely, if the fiscal rule is aggressive and consolidates quickly by reducing transfers when debt rises, the additional demand created by monetary easing is partly neutralized.

This interaction implies that in a high-inequality environment, the macroeconomic consequences of monetary policy depend more heavily on the fiscal regime. Coordinated policies, in which the central bank and the fiscal authority jointly internalize the distribution of PLI and the associated patterns of MPC and EIS, can exploit the strengthened multipliers to stabilize output more effectively. Uncoordinated policies, where fiscal policy leans against monetary stimulus or tightens in response to higher debt, can significantly dampen the effectiveness of monetary policy. Consequently, the rise in PLI inequality reinforces the importance of fiscal–monetary coordination for effective stabilization.

Conclusion

This paper shows that rising permanent labor-income inequality has first-order implications for the transmission of monetary and fiscal policy. Once preferences feature a non-homothetic taste for wealth, higher permanent labor income inequality raises aggregate MPCs, lowers the effective elasticity of intertemporal substitution, and shifts monetary transmission toward income-driven channels. In equilibrium, these forces make monetary policy more powerful despite a weaker substitution channel, and they substantially amplify fiscal multipliers, increasing the scope for self-financing deficits. The results imply that fiscal–monetary interactions become increasingly important in high-inequality environments, as the output effects of monetary shocks hinge on the government’s fiscal response. More broadly, the analysis highlights that the structure of permanent-income inequality is a central determinant of stabilization policy in modern heterogeneous-agent economies.

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A Deriving the Intertemporal Keynesian Cross

A.1 Intertemporal Keynesian Cross following a monetary shock

In the goods space:

$$\mathbf{Y} = (1 - \omega)\mathbf{C}_h(\{r_s, \tilde{N}_s\}) + \omega\mathbf{C}_l(\{r_s, \tilde{N}_s\})$$

Differentiating:

$$d\mathbf{Y} = (1 - \omega)z_h\mathbf{M}_h^{\tilde{n}}d\tilde{\mathbf{N}} + (1 - \omega)z_h\mathbf{M}_h^r d\mathbf{r} + z_l\omega\mathbf{M}_l^{\tilde{n}}d\tilde{\mathbf{N}} + \omega z_l\mathbf{M}_l^r d\mathbf{r}$$

In general equilibrium, $d\mathbf{Y} = d\mathbf{N} = d\mathbf{C}$, $d\tau = Bd\mathbf{r} - rBd\mathbf{N}$, $\tau = rB$ and $d\tilde{\mathbf{N}} = (1 - \tau)d\mathbf{N} - d\tau$.

Hence, $d\tilde{\mathbf{N}} = (1 - rB)d\mathbf{Y} - Bd\mathbf{r} + rBd\mathbf{Y} = d\mathbf{Y} - Bd\mathbf{r}$

We have that

$$d\mathbf{Y} = (1 - \omega)z_h\mathbf{M}_h^{\tilde{n}}(d\mathbf{Y} - Bd\mathbf{r}) + (1 - \omega)z_h\mathbf{M}_h^r d\mathbf{r} + z_l\omega\mathbf{M}_l^{\tilde{n}}(d\mathbf{Y} - Bd\mathbf{r}) + \omega z_l\mathbf{M}_l^r d\mathbf{r}$$

Solving for $d\mathbf{Y}$:

$$(\mathbf{I} - (1 - \omega)z_h\mathbf{M}_h^{\tilde{n}} - z_l\omega\mathbf{M}_l^{\tilde{n}}) d\mathbf{Y} = ((1 - \omega)z_h\mathbf{M}_h^r + \omega z_l\mathbf{M}_l^r d\mathbf{r} - B(1 - \omega)z_h\mathbf{M}_h^{\tilde{n}} - Bz_l\omega\mathbf{M}_l^{\tilde{n}}) d\mathbf{r}$$

Using that $\forall x \in \{r, \tilde{n}\}$, $\mathbf{M}^x = \omega z_l\mathbf{M}_l^x + (1 - \omega)z_h\mathbf{M}_h^x$,

$$(\mathbf{I} - \mathbf{M}^{\tilde{n}}) d\mathbf{Y} = (\mathbf{M}^r - B\mathbf{M}^{\tilde{n}}) d\mathbf{r}$$

Mutiplied by $\mathbf{K} \equiv -\sum_{t=1}^{\infty} (1 + r)^{-t} \mathbf{F}^t$ on both sides with $\mathbf{F}$ the forward matrix. Then inverting, we have that:

$$d\mathbf{Y} = (\mathbf{K} (\mathbf{I} - \mathbf{M}^{\tilde{n}}))^{-1} \mathbf{K} (\mathbf{M}^r - B\mathbf{M}^{\tilde{n}}) d\mathbf{r}$$

A.2 Decomposition of the effect of a change in the permanent labor income distribution

In a economy characterized by a permanent labor income distribution, the effect of a monetary shock $d\mathbf{r}$ is given by

$$d\mathbf{Y}(\mathbf{z}) = ((1 - \omega)z_h\mathbf{M}_h^r(\mathbf{z}) + \omega z_l\mathbf{M}_l^r(\mathbf{z}))d\mathbf{r} + ((1 - \omega)z_h\mathbf{M}_h^n(\mathbf{z}) + \omega z_l\mathbf{M}_l^n(\mathbf{z}))d\tilde{\mathbf{N}}(\mathbf{z})$$

Taking the difference between the effect of the same monetary shock in two economies characterized by two different permanent income distribution $\mathbf{z}'$ and $\mathbf{z}$:

$$\begin{aligned} d\mathbf{Y}(\mathbf{z}') - d\mathbf{Y}(\mathbf{z}) = & ((1 - \omega)z'_h\mathbf{M}_h^r(\mathbf{z}') + \omega z'_l\mathbf{M}_l^r(\mathbf{z}'))d\mathbf{r} + ((1 - \omega)z'_h\mathbf{M}_h^n(\mathbf{z}') + \omega z'_l\mathbf{M}_l^n(\mathbf{z}'))d\tilde{\mathbf{N}}(\mathbf{z}') \\ & - ((1 - \omega)z_h\mathbf{M}_h^r(\mathbf{z}) + \omega z_l\mathbf{M}_l^r(\mathbf{z}))d\mathbf{r} - ((1 - \omega)z_h\mathbf{M}_h^n(\mathbf{z}) + \omega z_l\mathbf{M}_l^n(\mathbf{z}))d\tilde{\mathbf{N}}(\mathbf{z}) \end{aligned}$$

$$\begin{aligned} d\mathbf{Y}(\mathbf{z}') - d\mathbf{Y}(\mathbf{z}) = & ((1 - \omega)(z'_h - z_h)\mathbf{M}_h^r(\mathbf{z}') + \omega(z'_l - z_l)\mathbf{M}_l^r(\mathbf{z}'))d\mathbf{r} \\ & + ((1 - \omega)(z'_h - z_h)\mathbf{M}_h^n(\mathbf{z}') + \omega(z'_l - z_l)\mathbf{M}_l^n(\mathbf{z}'))d\tilde{\mathbf{N}}(\mathbf{z}') \\ & + ((1 - \omega)z_h(\mathbf{M}_h^r(\mathbf{z}') - \mathbf{M}_h^r(\mathbf{z})) + \omega z_l(\mathbf{M}_l^r(\mathbf{z}') - \mathbf{M}_l^r(\mathbf{z})))d\mathbf{r} \\ & + ((1 - \omega)z_h(\mathbf{M}_h^n(\mathbf{z}') - \mathbf{M}_h^n(\mathbf{z})) + \omega z_l(\mathbf{M}_l^n(\mathbf{z}') - \mathbf{M}_l^n(\mathbf{z})))d\tilde{\mathbf{N}}(\mathbf{z}) \\ & + \mathbf{M}^{\tilde{n}}(\mathbf{z})(d\tilde{\mathbf{N}}(\mathbf{z}') - d\tilde{\mathbf{N}}(\mathbf{z})) \end{aligned}$$

Defining $\Delta z_i \equiv z'_i - z_i$ and $\Delta \mathbf{M}^x(\mathbf{z}') \equiv \mathbf{M}^x(\mathbf{z}') - \mathbf{M}^x(\mathbf{z})$, we get

$$\begin{aligned} d\mathbf{Y}(\mathbf{z}') - d\mathbf{Y}(\mathbf{z}) = & [(1 - \omega)\Delta z'_h\mathbf{M}_h^r(\mathbf{z}') + \omega\Delta z'_l\mathbf{M}_l^r(\mathbf{z}')]d\mathbf{r} + [(1 - \omega)\Delta z'_h\mathbf{M}_h^{\tilde{n}}(\mathbf{z}') + \omega\Delta z'_l\mathbf{M}_l^{\tilde{n}}(\mathbf{z}')]d\tilde{\mathbf{N}}(\mathbf{z}') \\ & + [(1 - \omega)z_h\Delta \mathbf{M}_h^r(\mathbf{z}') + \omega z_l\Delta \mathbf{M}_l^r(\mathbf{z}')]d\mathbf{r} + [(1 - \omega)z_h\Delta \mathbf{M}_h^{\tilde{n}}(\mathbf{z}') + \omega z_l\Delta \mathbf{M}_l^{\tilde{n}}(\mathbf{z}')]d\tilde{\mathbf{N}}(\mathbf{z}') \\ & + \mathbf{M}^{\tilde{n}}(\mathbf{z})(d\tilde{\mathbf{N}}(\mathbf{z}') - d\tilde{\mathbf{N}}(\mathbf{z})). \end{aligned}$$

B Analytical Approximation of Pareto Tails in the Model

Consider a household of permanent type i solving

$$V_i(a_{t-1}, e_t) = \max_{c_t, a_t \geq 0} \frac{c_t^{1-\sigma} - 1}{1-\sigma} + \gamma \frac{\left(\frac{a_t}{A_t} + \zeta\right)^{1-\Sigma}}{1-\Sigma} + \beta(1-\zeta) \mathbb{E}_t[V_i(a_t, e_{t+1})],$$

subject to

$$c_t + a_t = (1 - \tau_t)(z_i e_t N_t)^{1-\phi} + (1 + r_t)a_{t-1} + \chi_t + \psi_t,$$

and $a_t \geq 0$. Since the derivation focuses on the steady state, we assume $r_t = r$, $A_t = A$, and abstract from transfers and inheritance ($\chi_t = \psi_t = 0$). The proof trivially extends when $\chi_t, \psi_t > 0$ as long as they are not correlated with individual wealth.

The Euler equation is

$$c_t^{-\sigma} = (1 - \zeta)\beta(1 + r)\mathbb{E}_t[c_{t+1}^{-\sigma}] + \frac{\gamma}{A} \left(\frac{a_t}{A} + \zeta\right)^{-\Sigma}.$$

For large wealth $a_{t-1} \rightarrow \infty$, labor income becomes negligible and idiosyncratic shocks vanish from the limiting policy function. The problem simplifies to

$$V(a_{t-1}) = \frac{c_t^{1-\sigma}}{1-\sigma} + \gamma \frac{(a_t/A + \zeta)^{1-\Sigma}}{1-\Sigma} + \beta(1-\zeta)V(a_t),$$

subject to

$$c_t + a_t = (1 + r)a_{t-1}.$$

The Euler equation thus becomes

$$c_t^{-\sigma} = (1 - \zeta)\beta(1 + r)c_{t+1}^{-\sigma} + \frac{\gamma}{A} \left(\frac{a_t}{A} + \zeta\right)^{-\Sigma}.$$

We guess a power-law consumption function of the form

$$c_t = B a_{t-1}^{\Sigma/\sigma}.$$

Then the asset accumulation equation becomes

$$a_t = (1 + r)a_{t-1} - B a_{t-1}^{\Sigma/\sigma}.$$

As $a_{t-1} \rightarrow \infty$,

$$a_t \sim (1 + r)a_{t-1}.$$

Thus, asymptotically,

$$c_{t+1} = B a_t^{\Sigma/\sigma} \sim B(1 + r)^{\Sigma/\sigma} a_{t-1}^{\Sigma/\sigma}.$$

Plugging into the Euler equation:

$$B^{-\sigma} a_{t-1}^{-\Sigma} = (1 - \xi)\beta(1 + r) B^{-\sigma} (1 + r)^{-\Sigma} a_{t-1}^{-\Sigma} + \frac{\gamma}{A} \left(\frac{a_{t-1}}{A} \right)^{-\Sigma}.$$

Dividing both sides by $a_{t-1}^{-\Sigma}$ and rearranging gives:

$$B^{-\sigma} = (1 - \xi)\beta(1 + r)^{1-\Sigma} B^{-\sigma} + \frac{\gamma}{A} \left(\frac{1}{A} \right)^{-\Sigma}.$$

Solving for B yields the scalar nonlinear equation:

$$B = \left[(1 - \xi)\beta(1 + r)^{1-\Sigma} B^{-\sigma} + \frac{\gamma}{A^{1-\Sigma}} \right]^{-1/\sigma},$$

which can be solved numerically. The law of motion for wealth at very high levels is

$$a_t \approx (1 + r)a_{t-1}.$$

With death probability ξ , the transition is:

1. with probability $(1 - \xi)$: wealth grows by $(1 + r)$
2. with probability ξ : wealth becomes 0 (death and restart at low wealth)

Thus, the multiplicative growth factor is

$$G_t = \begin{cases} 1 + r, & \text{with prob. } (1 - \xi), \\ 0, & \text{with prob. } \xi. \end{cases}$$

Let $s = 0$ denote "alive" and $s = 1$ denote "dead". The conditional moment generating matrix is:

$$M(z) = \begin{pmatrix} (1+r)^z & 0 \\ (1+r)^z & 0 \end{pmatrix}.$$

The survival transition matrix is

$$P = \begin{pmatrix} 1 - \xi & \xi \\ 1 - \xi & \xi \end{pmatrix}.$$

The Beare–Toda condition for a Pareto tail is

$$\rho(P \circ M(z)) = 1,$$

where $\circ$ is the Hadamard product.

Thus

$$P \circ M(z) = \begin{pmatrix} (1 - \xi)(1+r)^z & 0 \\ (1 - \xi)(1+r)^z & 0 \end{pmatrix},$$

whose spectral radius is

$$\rho = (1 - \xi)(1+r)^z.$$

Setting $\rho = 1$ yields the Pareto exponent:

$$(1 - \xi)(1+r)^\xi = 1,$$

so

$$\boxed{\xi^a = \frac{-\log(1 - \xi)}{\log(1 + r)}}.$$

From the asymptotic consumption rule,

$$c_t = Ba_{t-1}^{\Sigma/\sigma},$$

we invert:

$$a_{t-1} = B^{-\sigma/\Sigma} c_t^{\sigma/\Sigma}.$$

Since a has a Pareto tail with parameter ζ , i.e.

$$P(a > x) \sim Cx^{-\zeta},$$

we obtain

$$P(c > y) = P\left(a > B^{-\sigma/\Sigma} y^{\sigma/\Sigma}\right) \sim C \left(B^{-\sigma/\Sigma} y^{\sigma/\Sigma}\right)^{-\zeta},$$

so

$$P(c > y) \sim C' y^{-\zeta\sigma/\Sigma}.$$

Thus consumption has a Pareto tail with exponent

$$\boxed{\zeta^c = \zeta^a \frac{\sigma}{\Sigma}}.$$